MACHINE LEARNING

Module Code: CSC83101

LECTURE

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Prerequisites

- This course assumes that you are comfortable programming in Python and that you are familiar with Python's main scientific libraries, in particular NumPy, Pandas, and Matplotlib
- Experience with notebook environments
- Understanding of college-level math as well (calculus, linear algebra, probabilities, and statistics)

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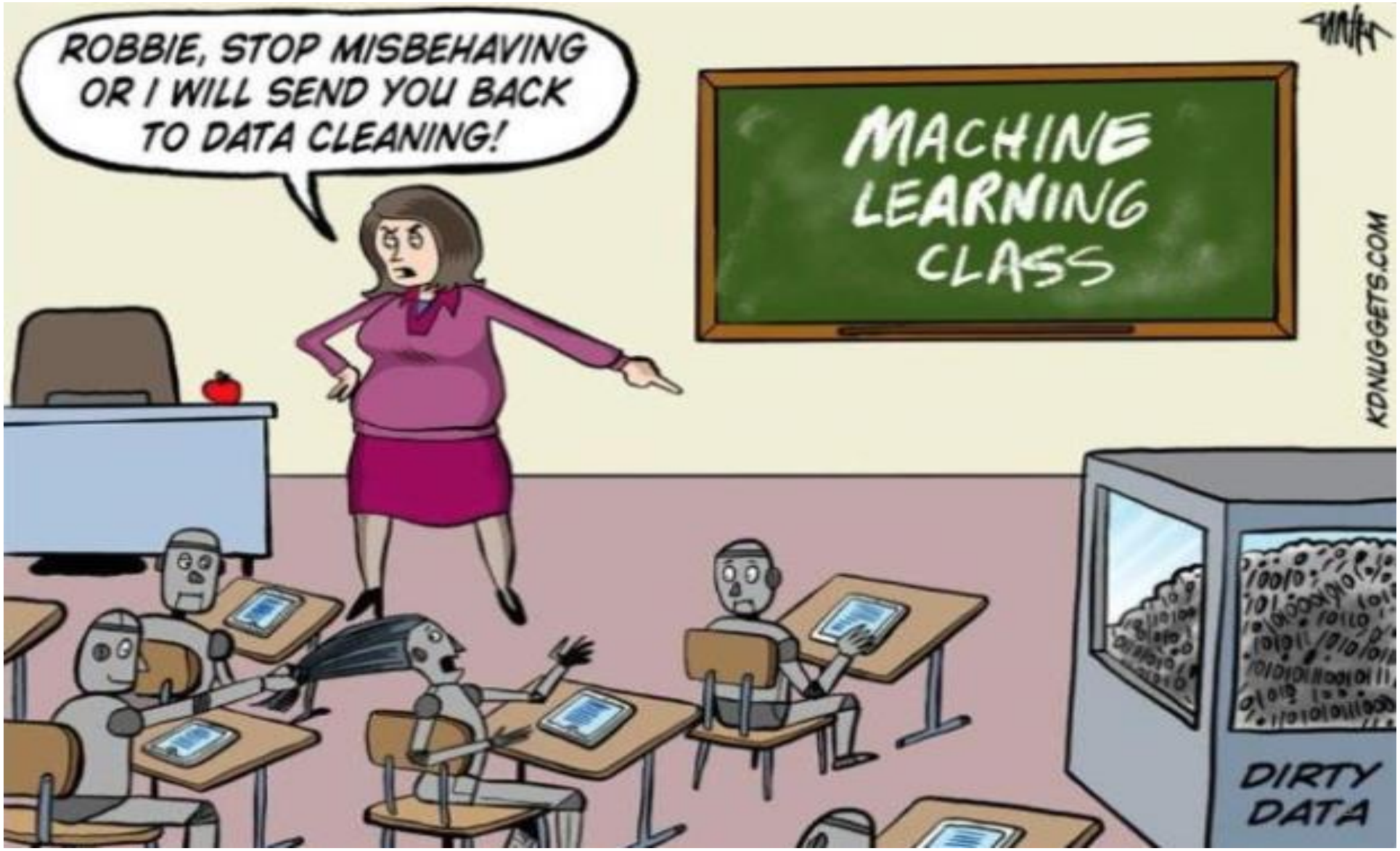

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Unit 1: Foundations of Machine Learning and Python Programming

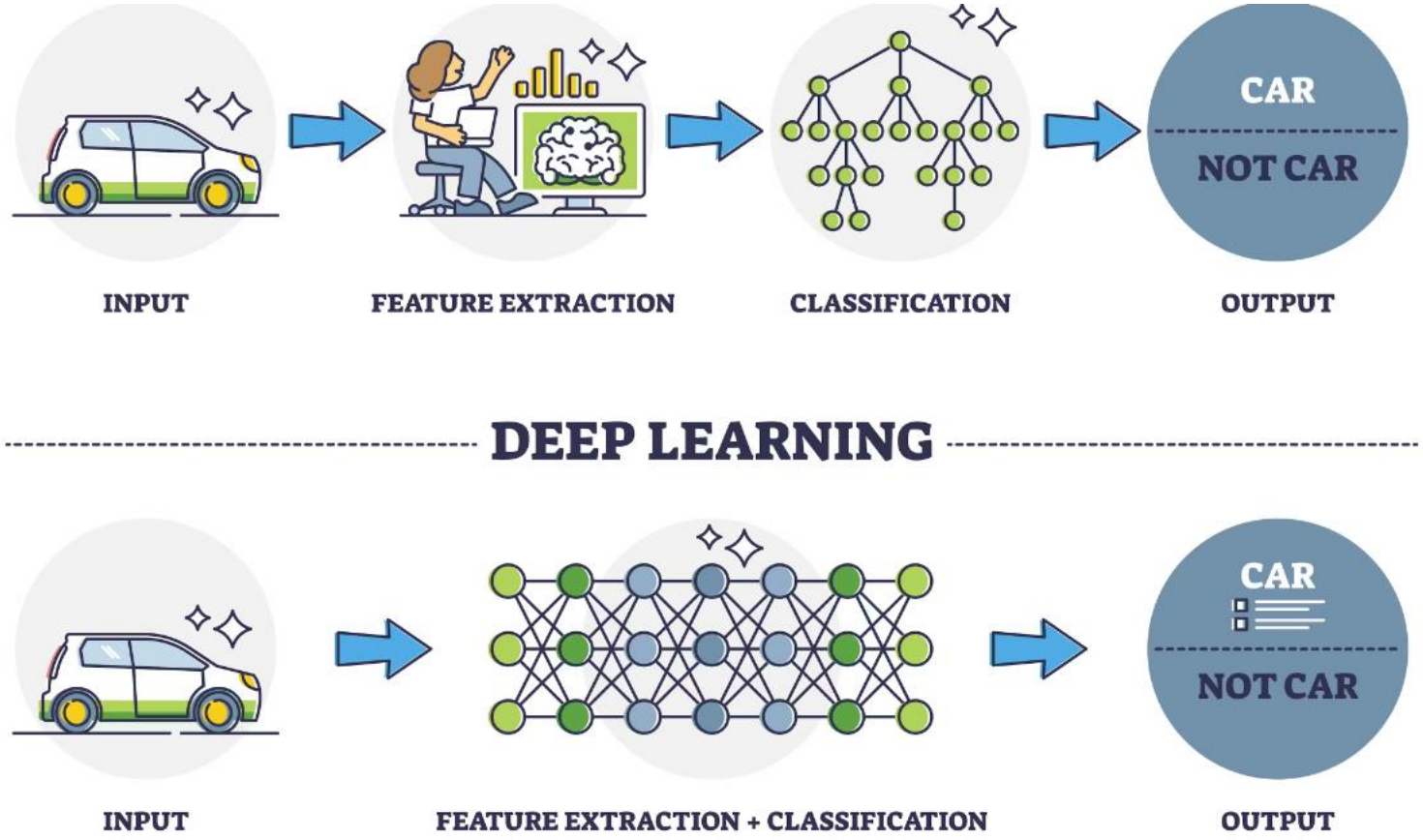
- **Objective:**

Build foundational knowledge of machine learning concepts and essential Python programming skills needed for practical ML implementation.

Indicative Contents:

- Introduction to Machine Learning
  - Definition, scope, and applications
  - Types of learning: supervised, unsupervised, reinforcement learning
- Python Programming for ML
  - Basic Python syntax and data structures
  - Control flow (loops, conditionals)
  - Functions and modules
  - Using libraries: NumPy, pandas, matplotlib/seaborn
  - Data input/output (CSV, JSON)
- Data Preprocessing and Feature Engineering
  - Handling missing data and outliers
  - Data normalization and scaling
  - Encoding categorical variables
  - Feature selection and extraction

Introduction to Machine Learning



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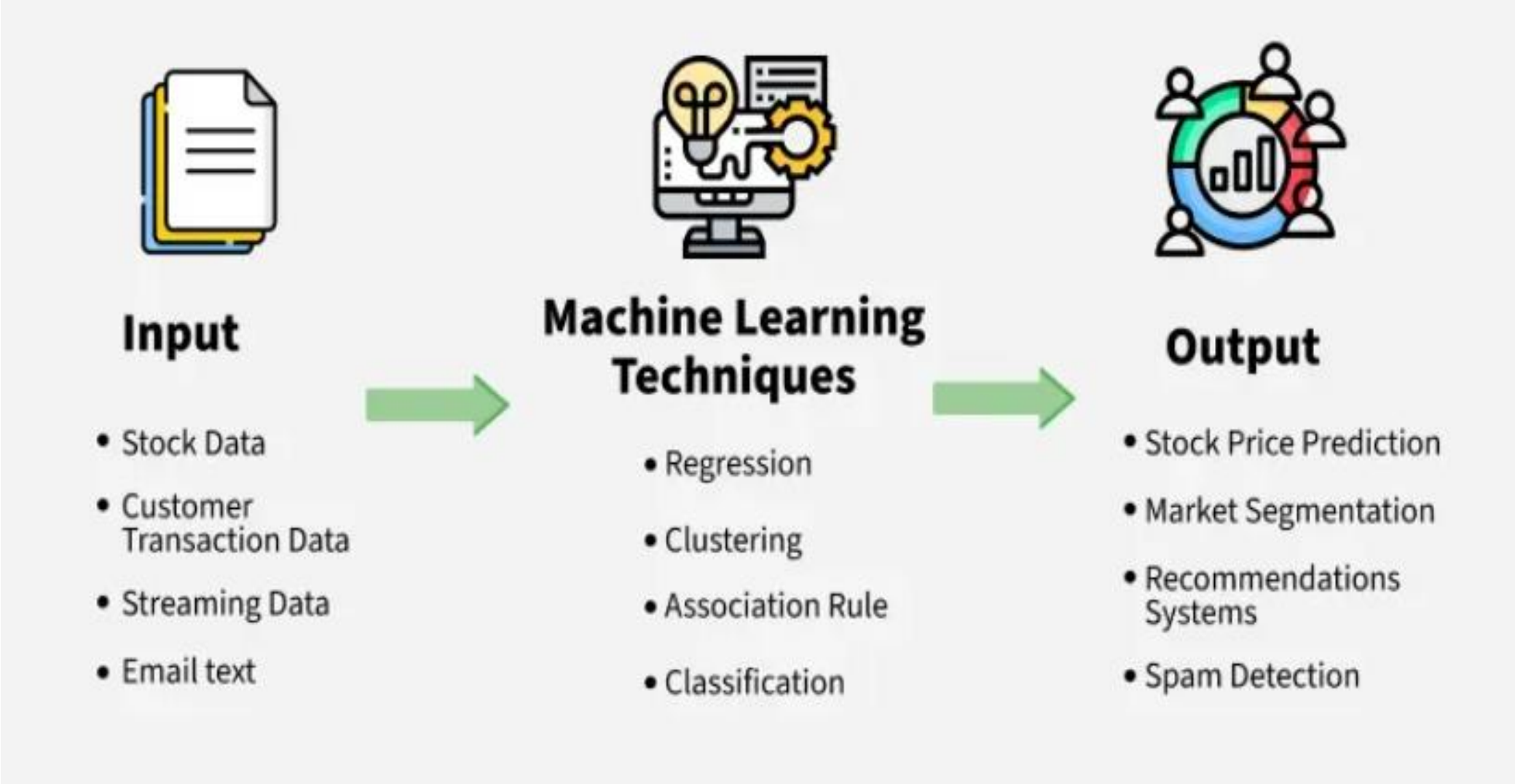
Introduction to Machine

Arthur Samuel, an early American leader in the field of computer gaming and artificial intelligence, coined the term "*Machine learning*" in 1959 while at IBM. He defined machine learning as "the field of study that gives computers the ability to learn without being explicitly programmed."



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What is Machine Learning?

- Machine Learning is a concept which allows the machine to learn from examples and experience, and that too without being explicitly programmed.
- So instead of you writing the code, what you do is you feed data to the generic algorithm, and the algorithm/ machine builds the logic based on the given data

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Machine learning (ML)

Machine learning (ML) is a subset of **artificial intelligence (AI)** focused on building systems that learn and improve from data without being explicitly programmed for specific tasks. While traditional programming uses fixed, hand-coded rules, ML uses statistical algorithms to discover patterns and make predictions autonomously.

How does Machine Learning Work?

- Machine Learning algorithm is trained using a training data set to create a model. When new input data is introduced to the ML algorithm, it makes a prediction on the basis of the model.
- The prediction is evaluated for accuracy and if the accuracy is acceptable, the Machine Learning algorithm is deployed. If the accuracy is not acceptable, the Machine Learning algorithm is trained again and again with an augmented raining data set

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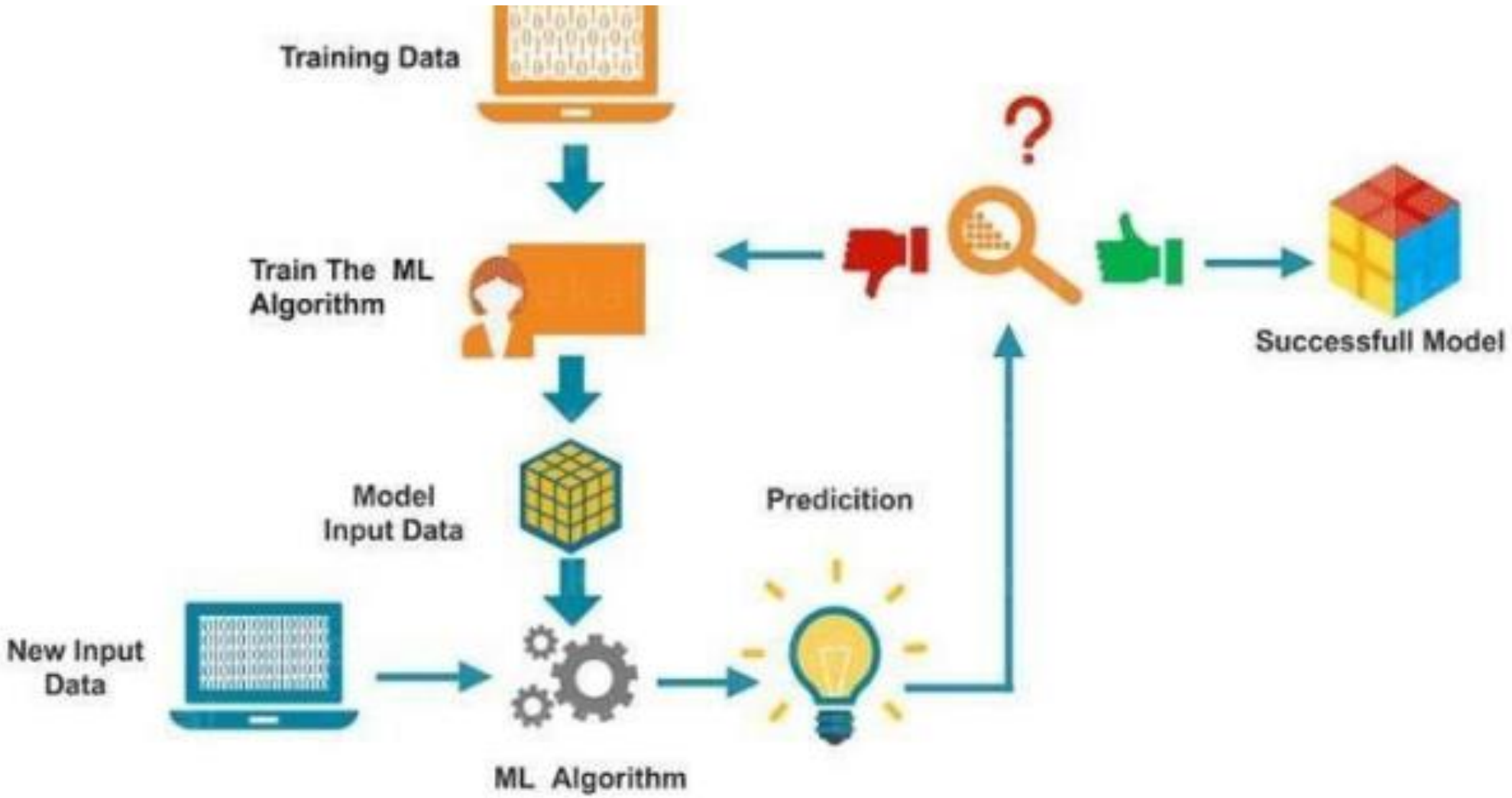

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Machine learning algorithms

- *Machine learning algorithms* - computer programs that adjust themselves to perform better as they are exposed to more data.
- The "learning" part of machine learning means these programs change how they process data over time.
- In other words, a machine learning algorithm can adjust its own settings, given feedback on its previous performance in making predictions about a collection of data (images, text, etc.)

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- A computer program is said to *learn* from experience *E* with respect to some class of tasks *T* and performance measure *P*, if its performance at tasks *T*, as measured by *P*, improves with experience *E*.
- Examples

1. Handwriting recognition learning problem.

- Task T : Recognising and classifying handwritten words within images
- Performance P : Percent of words correctly classified
- Training experience E: A dataset of handwritten words with given classifications

2. A robot driving learning problem

- Task T : Driving on highways using vision sensors
- Performance measure P : Average distance traveled before an error
- training experience: A sequence of images and steering commands recorded while observing a human driver

3. A chess learning problem

- Task T : Playing chess
- Performance measure *P* : Percent of games won against opponents
- Training experience *E*: Playing practice games against itself

2. Basic components of learning process

The learning process, whether by a human or a machine, can be divided into four components, namely, data storage, abstraction, generalization and evaluation. Figure 1.1 illustrates the various components and the steps involved in the learning process.

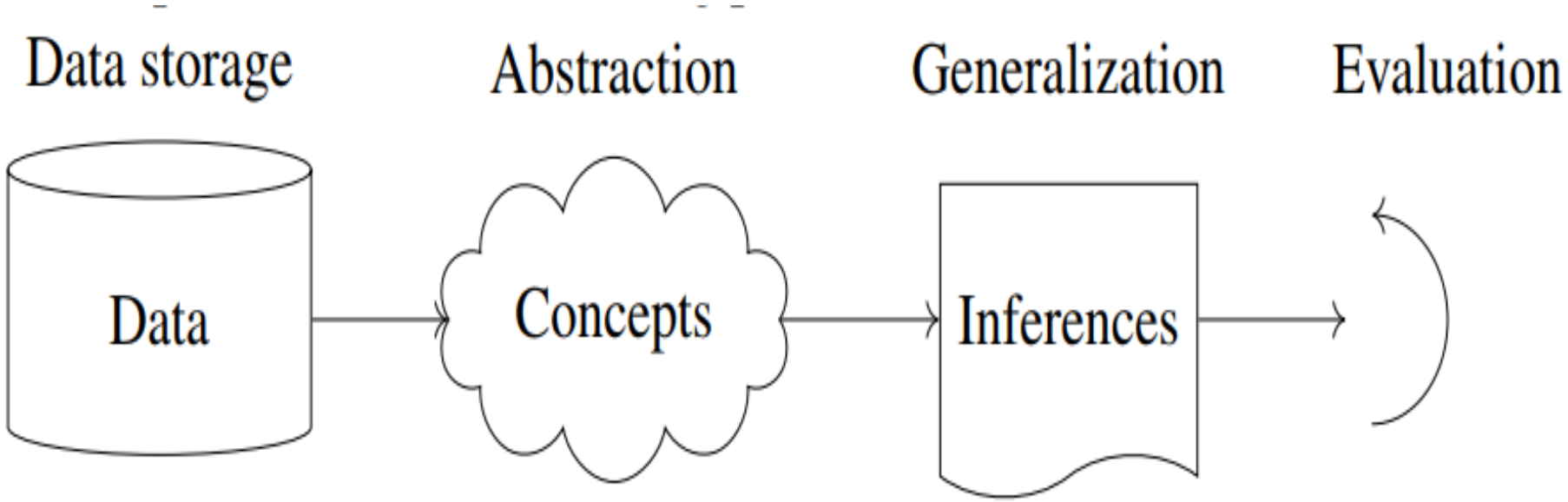

Figure 1.1: Components of learning process

Data storage

- Facilities for storing and retrieving huge amounts of data are an important component of the learning process. Humans and computers alike utilize data storage as a foundation for advanced reasoning.

- In a human being, the data is stored in the brain and data is retrieved using electrochemical signals.

- Computers use hard disk drives, flash memory, random access memory and similar devices to store data and use cables and other technology to retrieve data.

2. Abstraction

- The second component of the learning process is known as *abstraction*.
- Abstraction is the process of extracting knowledge about stored data. This involves creating general concepts about the data as a whole.
- The creation of knowledge involves application of known models and creation of new models.
- The process of fitting a model to a dataset is known as *training*. When the model has been trained, the data is transformed into an abstract form that summarizes the original information.

3. Generalization

- The third component of the learning process is known as *generalization.*

- The term generalization describes the process of turning the knowledge about stored data into a form that can be utilized for future action. These actions are to be carried out on tasks that are similar, but not identical, to those what have been seen before. In generalization, the goal is to discover those properties of the data that will be most relevant to future tasks.

4. Evaluation

- *Evaluation* is the last component of the learning process.
- It is the process of giving feedback to the user to measure the utility of the learned knowledge. This feedback is then utilised to effect improvements in the whole learning process.

Why Machine Learning

Data
44 Zettabytes by 2020*

Cloud
Cheap, scalable compute

Data Science
Data scientist is the sexiest job**

Internet of Things
212B connected things by 2020*

Storage
Access to cheap, abundant storage

Algorithms
Open Source and democratized

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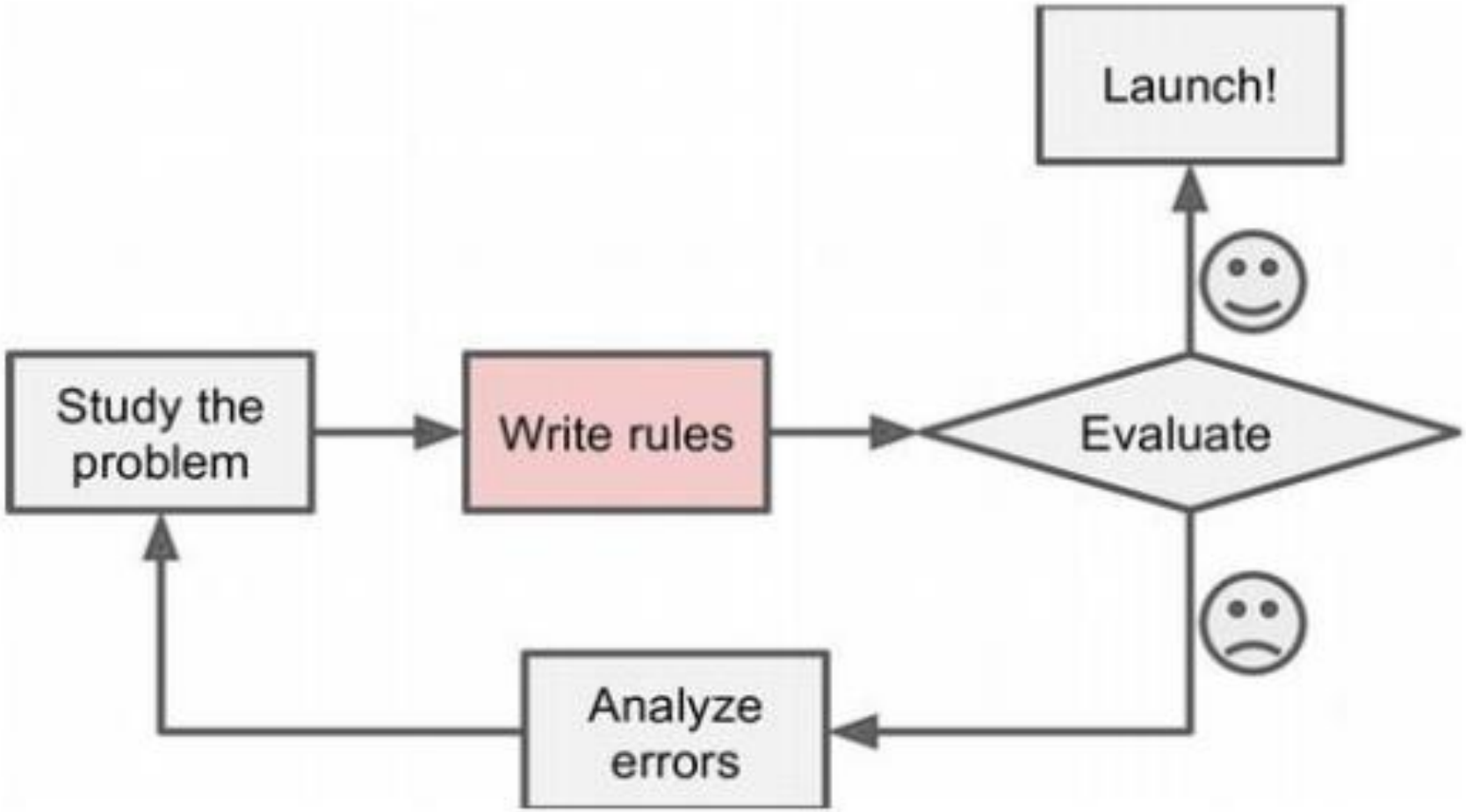
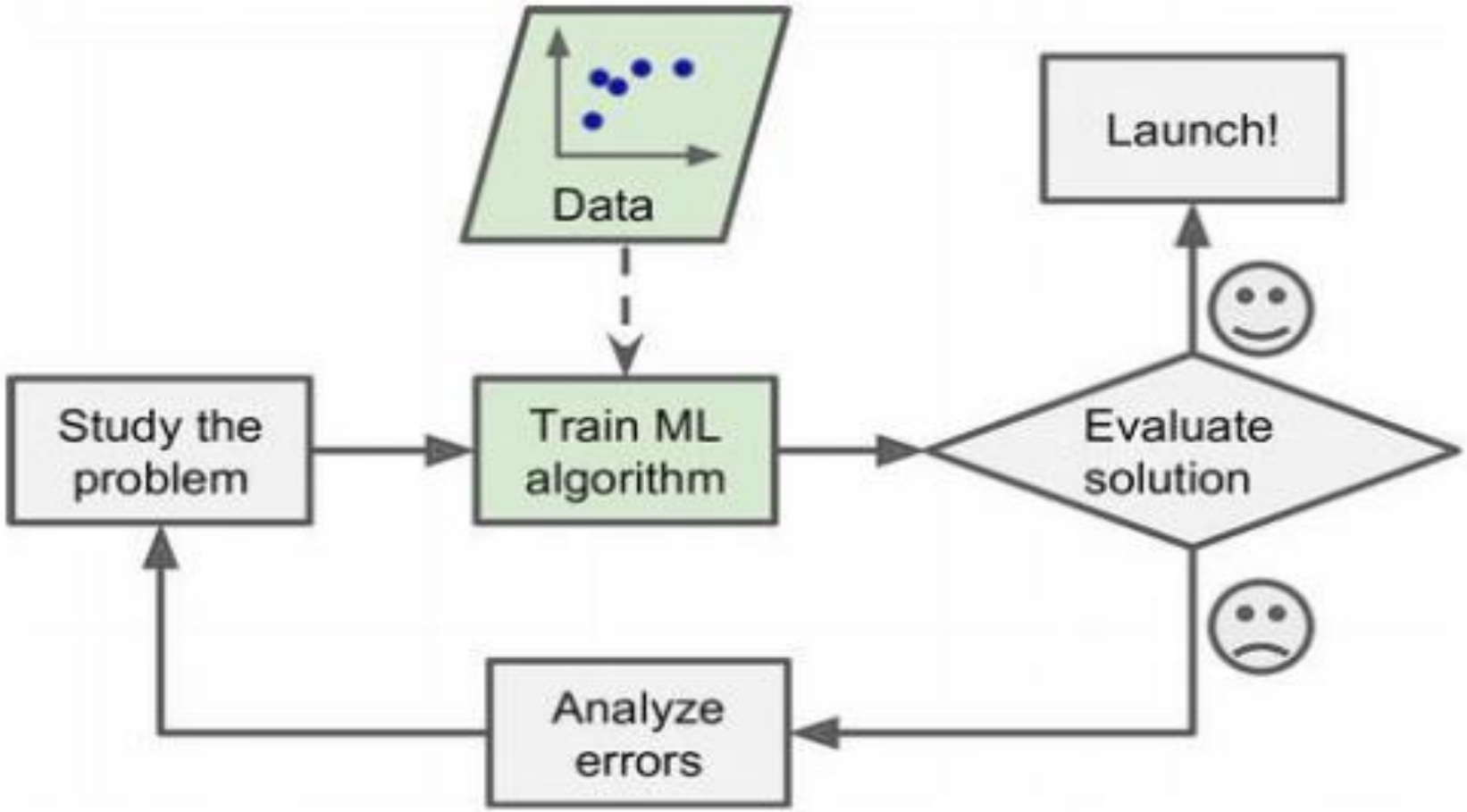
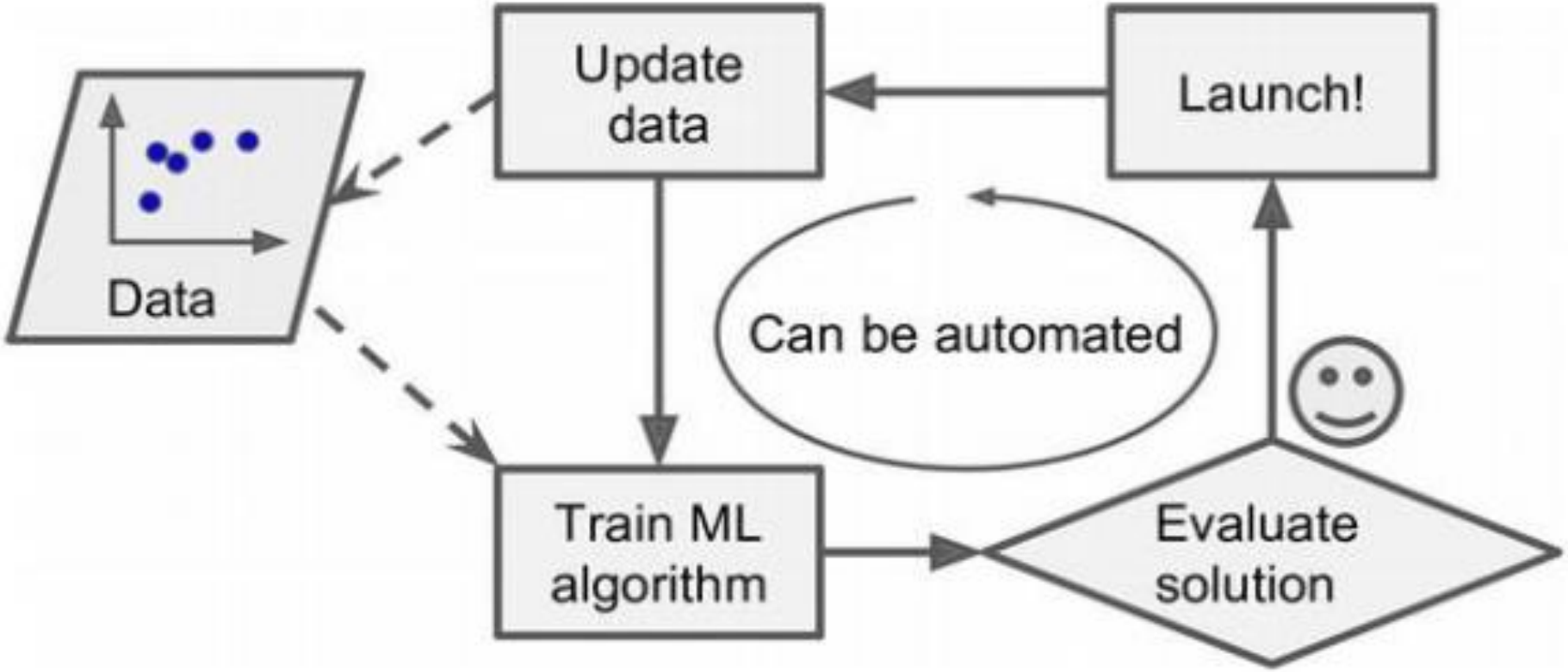
Traditional Approach



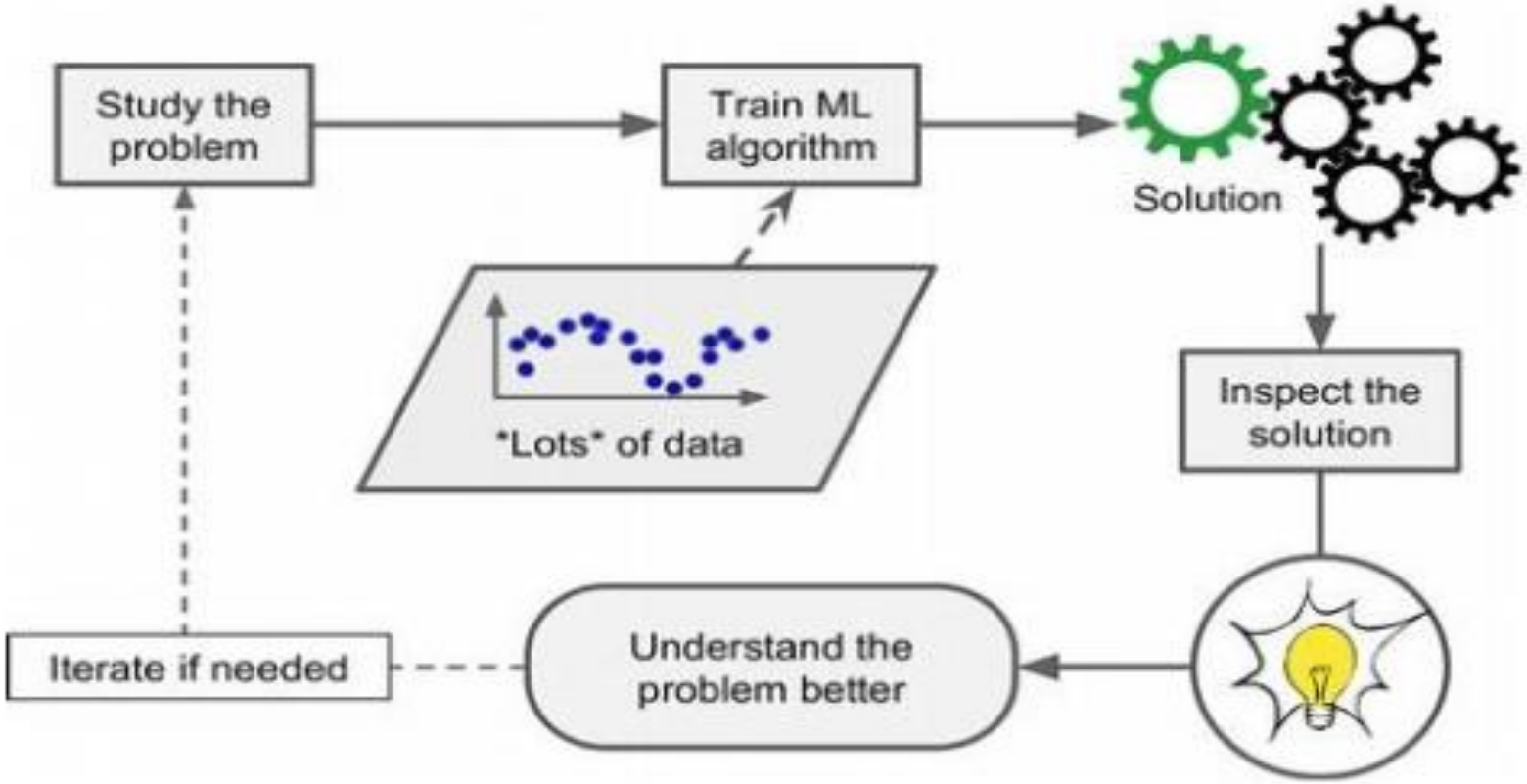
Machine Learning Approach



Machine Learning Automation



Machine Learning Automation (cont'd)



Application of machine learning

- **Healthcare and Medical Diagnosis:** ML models analyze medical images (X-rays, MRIs) to detect diseases like cancer or pneumonia. Predictive analytics are used to assess patient risks, personalized treatment plans, and accelerate drug discovery.
- **Finance and Banking:** Banks use ML for real-time **fraud detection** by identifying anomalies in transaction patterns. Other uses include credit scoring for loans, algorithmic trading, and stock market trend prediction.
- **E-commerce and Retail:** Platforms like Amazon and Netflix use recommendation engines to suggest products and content based on user history. Retailers also use ML for dynamic pricing and demand forecasting.

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- **Transportation and Autonomous Vehicles:** Self-driving cars, such as those by Tesla, use computer vision and reinforcement learning to navigate and avoid obstacles. Apps like Google Maps use ML to predict traffic conditions and optimize routes.
- **Personalized Digital Assistants:** Virtual assistants like Siri, Alexa, and Google Assistant rely on speech recognition and Natural Language Processing (NLP) to follow voice commands.
- **Email and Cybersecurity:** ML filters **spam and malware** by recognizing patterns in content and sender behavior. It is also used to detect and respond to cyberattacks in real-time.

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- **Manufacturing and Industrial Processes: Predictive maintenance** uses sensor data to forecast when machinery might fail, reducing downtime. ML also improves quality control through automated defect detection.

- **Social Media:** Platforms like Facebook and Instagram use ML for auto-tagging (facial recognition), personalized news feeds, and content moderation

Understanding Data in Machine Learning

- Data is the foundation of machine learning; the quality and quantity of data directly impact the model's performance.

- Machine learning algorithms learn patterns, make decisions, or predictions based on data.

Types of Data in Machine Learning

- **1. Structured Data**
- Structured data refers to information that is highly organized and formatted in a way that is easily searchable and analyzable. This data is typically arranged in rows and columns within tables, such as in relational databases or spreadsheets. Examples include customer information (name, age, purchase history), sales records, and sensor readings. Because of its organized format, structured data can be readily processed using traditional data analysis tools and machine learning algorithms.

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Key Characteristics:

- Clearly defined schema (fields and data types)
- Easy to store, query, and manipulate
- Examples: Excel files, SQL databases, CSV files

2. Unstructured Data

- Unstructured data lacks a predefined format or organization, making it more complex to process and analyze. This type of data includes text documents, images, audio files, video recordings, social media posts, emails, and sensor data from IoT devices.
- Since there is no fixed structure, machine learning models often require significant preprocessing steps such as feature extraction, natural language processing (NLP), or image recognition techniques to make sense of unstructured data.

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Key Characteristics:

- No fixed schema or format
- Requires specialized techniques for analysis
- Examples: Text files, photos, videos, audio recordings

3. Semi-structured Data

- Semi-structured data lies between structured and unstructured data. It does not fit neatly into relational tables but contains tags or markers to separate semantic elements and enforce hierarchies of records and fields. This makes it more organized than unstructured data but less rigid than structured data. Common formats include JSON (JavaScript Object Notation), XML (eXtensible Markup Language), and YAML files, often used in web data, configuration files, and data interchange between systems.

Key Characteristics:

- Contains elements of structure through tags or keys
- Flexible and self-describing format
- Easier to parse than unstructured data but not as straightforward as structured data
- Examples: JSON files used in APIs, XML documents, HTML pages

Types of Machine Learning

Machine learning is sub-categorized to three types:

- Supervised Learning – Train Me!
- Unsupervised Learning – I am self sufficient in learning
- Reinforcement Learning – My life My rules! (Hit & Trial)

Supervised Machine Learning

- Supervised learning is a type of machine learning where a model learns from labelled data—meaning every input has a corresponding correct output. The model makes predictions and compares them with the true outputs, adjusting itself to reduce errors and improve accuracy over time. The goal is to make accurate predictions on new, unseen data.
- For example, a model trained on images of handwritten digits can recognise new digits it has never seen

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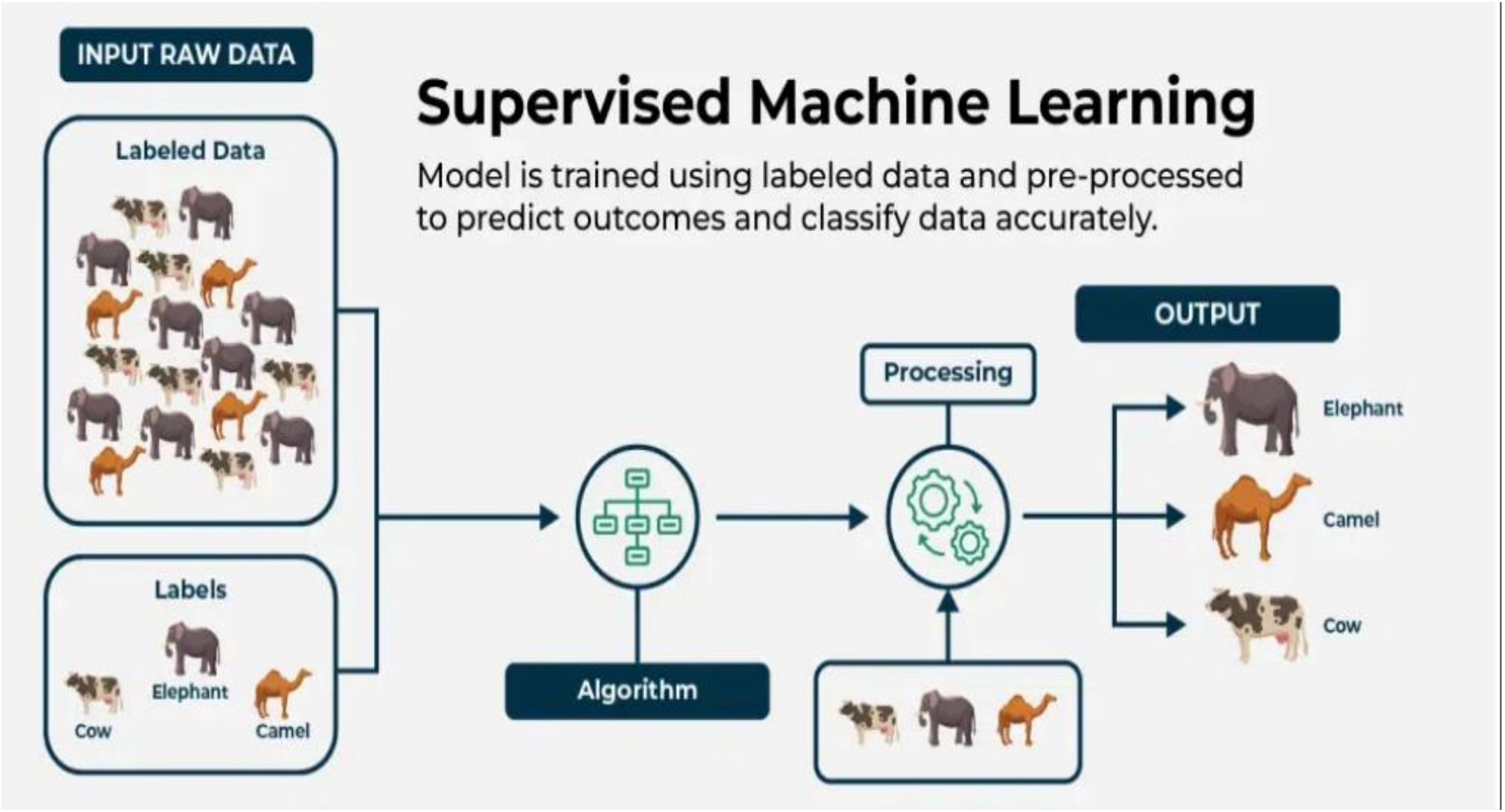
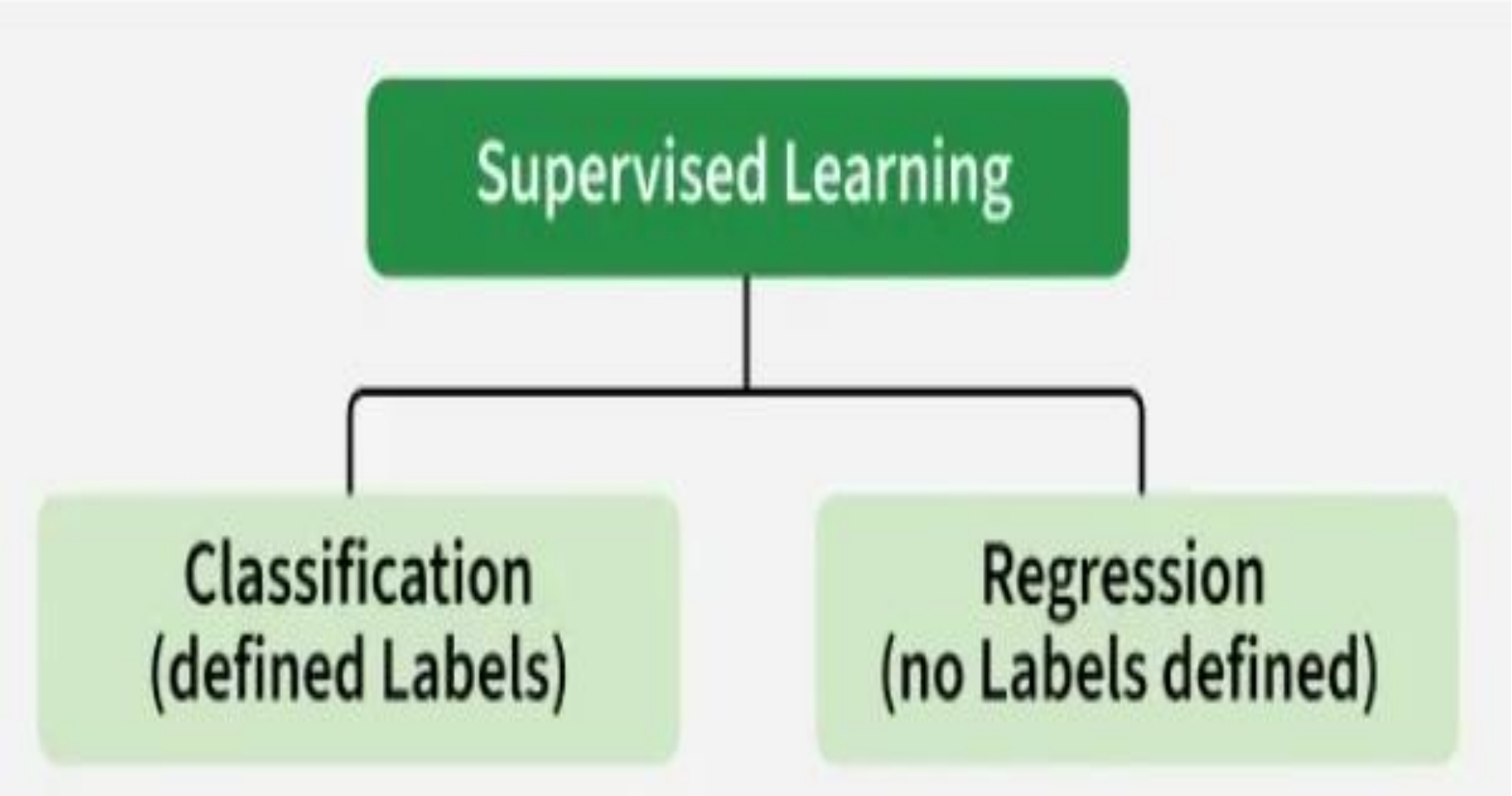

INPUT RAW DATA
Supervised Machine Learning
Model is trained using labeled data and pre-processed to predict outcomes and classify data accurately.
Labeled Data
Labels
Cow
Elephant
Camel
Algorithm
Processing
OUTPUT
Elephant
Camel
Cow

Types of Supervised Learning in Machine Learning

- **Classification**: Where the output is a categorical variable (e.g., spam vs. non-spam emails, yes vs. no).

- **Regression**: Where the output is a continuous variable (e.g., predicting house prices, stock prices).


Supervised Learning
Classification
(defined Labels)
Regression
(no Labels defined)

- While training the model, data is usually split in the ratio of 80:20 i.e. 80% as training data and the rest as testing data. In training data, we feed input as well as output for 80% of data.
- The model learns from training data only. We use different supervised learning algorithms (which we will discuss in detail in the next section) to build our model. Let's first understand the

User ID	Gender	Age	Salary	Purchased
15624510	Male	19	19000	0
15810944	Male	35	20000	1
15668575	Female	26	43000	0
15603246	Female	27	57000	0
15804002	Male	19	76000	1
15728773	Male	27	58000	1
15598044	Female	27	84000	0
15694829	Female	32	150000	1
15600575	Male	25	33000	1
15727311	Female	35	65000	0
15570769	Female	26	80000	1
15606274	Female	26	52000	0
15746139	Male	20	86000	1
15704987	Male	32	18000	0
15628972	Male	18	82000	0
15697686	Male	29	80000	0
15733883	Male	47	25000	1

Figure A: CLASSIFICATION

Temperature	Pressure	Relative Humidity	Wind Direction	Wind Speed
10.69261758	986.882019	54.19337313	195.7150879	3.278597116
13.59184184	987.8729248	48.0648859	189.2951202	2.909167767
17.70494885	988.1119385	39.11965597	192.9273834	2.973036289
20.95430404	987.8500366	30.66273218	202.0752869	2.965289593
22.9278274	987.2833862	26.06723423	210.6589203	2.798230886
24.04233986	986.2907104	23.46918024	221.1188507	2.627005816
24.41475295	985.2338867	22.25082295	233.7911987	2.448749781
23.93361956	984.8914795	22.35178837	244.3504333	2.454271793
22.68800023	984.8461304	23.7538641	253.0864716	2.418341875
20.56425726	984.8380737	27.07867944	264.5071106	2.318677425
17.76400389	985.4262085	33.54900114	280.7827454	2.343950987
11.25680746	988.9386597	53.74139903	68.15406036	1.650191426
14.37810685	989.6819458	40.70884681	72.62069702	1.553469896
18.45114201	990.2960205	30.85038484	71.70604706	1.005017161
22.54895853	989.9562988	22.81738811	44.66042709	0.264133632
24.23155922	988.796875	19.74790765	318.3214111	0.329656571

Figure B: REGRESSION

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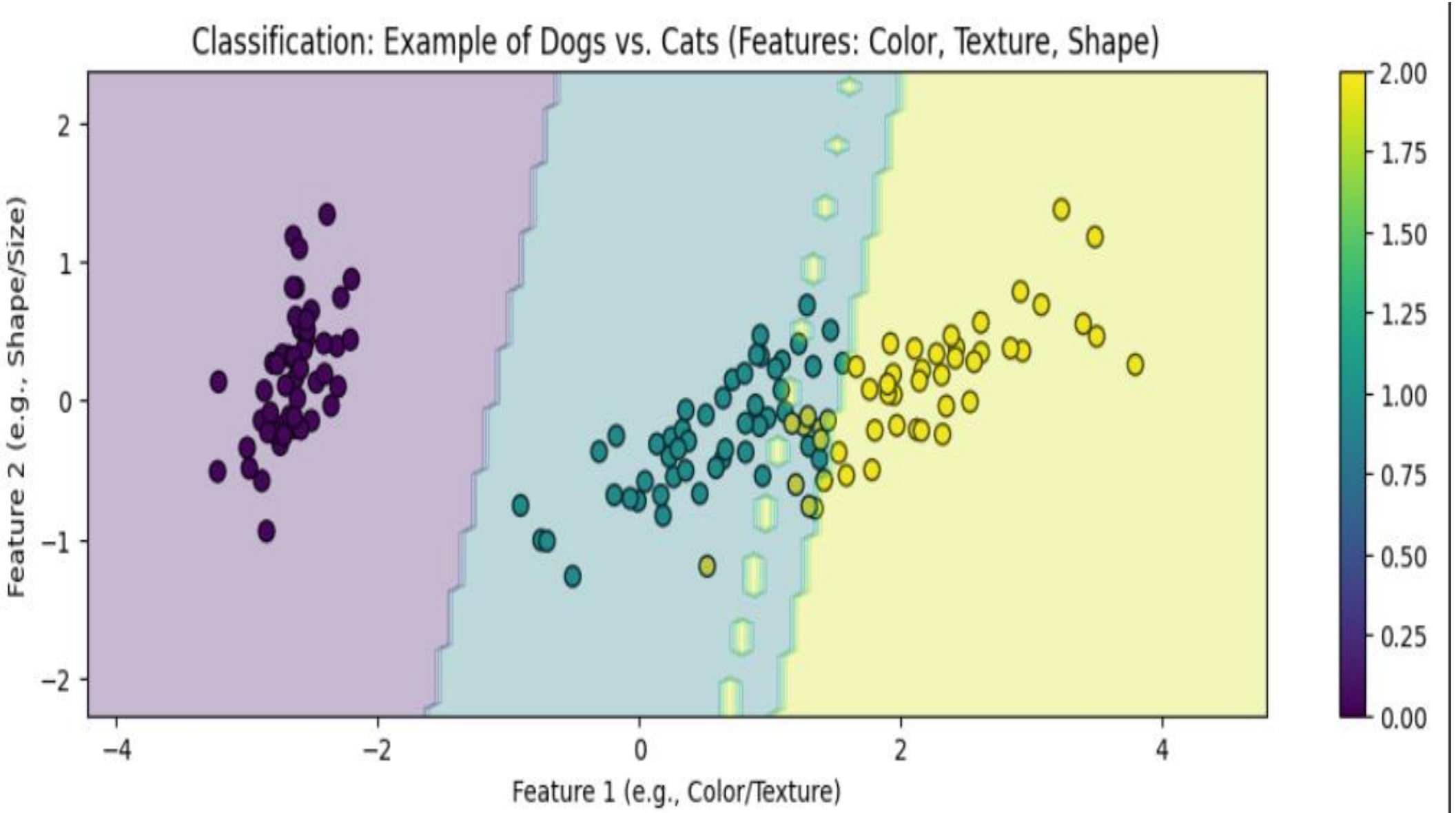
Getting started with Classification

- Classification is a supervised machine learning technique used to predict labels or categories based on input data. The goal is to assign each data point to a predefined class, such as spam vs. non-spam emails or diseased vs. healthy patients.
- For example, a classification model might be trained on dataset of images labeled as either dogs or cats and it can be used to predict the class of new and unseen images as dogs or cats based on their features such as colour, texture or shape.



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- Explaining classification in ML, the horizontal axis represents the combined values of colour and texture features.
- The vertical axis represents the combined values of shape and size features.
- Each coloured dot in the plot represents an individual image, with the colour indicating whether the model predicts the image to be a dog or a cat.
- The shaded areas in the plot show the decision boundary, which is the line or region that the model uses to decide which category (dog or cat) an image belongs to. The model classifies images on one side of the boundary as dogs and on the other side as cats, based on their features.

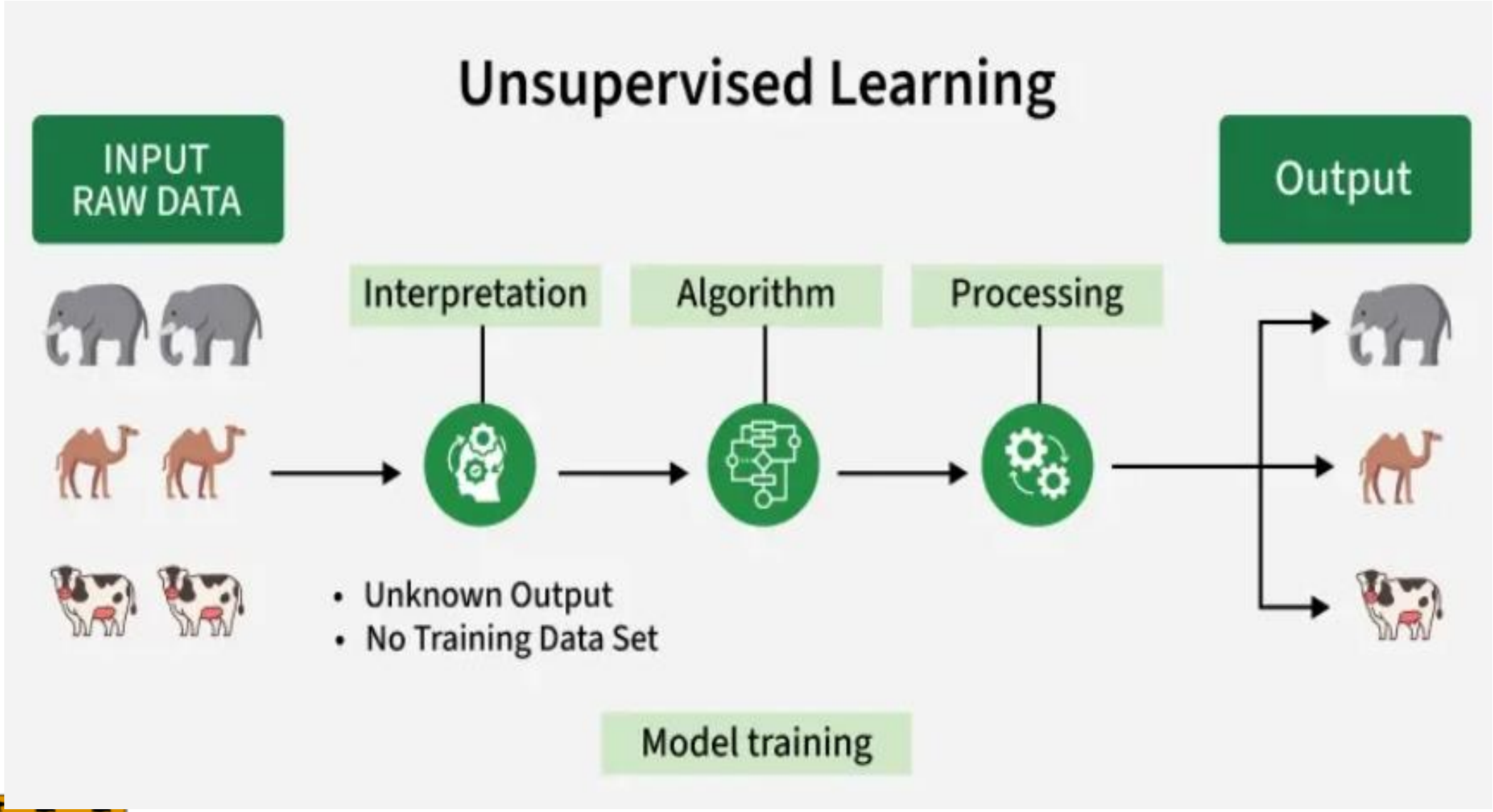
Regression in machine learning

- Regression in machine learning is a supervised learning technique used to predict continuous numerical values by learning relationships between input variables (features) and an output variable (target). It helps understand how changes in one or more factors influence a measurable outcome and is widely used in forecasting, risk analysis, decision-making and trend estimation

- Works with real-valued output variables
- Helps to identify strengths and the type of relationships
- Supports both simple and complex predictive models.
- Used for tasks like price prediction, trend forecasting and risk scoring.

What is Unsupervised Learning

- Unsupervised Learning is a type of machine learning where the model works without labelled data. It learns patterns on its own by grouping similar data points or finding hidden structures without any human intervention.
- It is used for tasks like clustering, dimensionality reduction and Association Rule Learning.
- Helps identify hidden patterns in data
- Useful for grouping, compression and anomaly detection

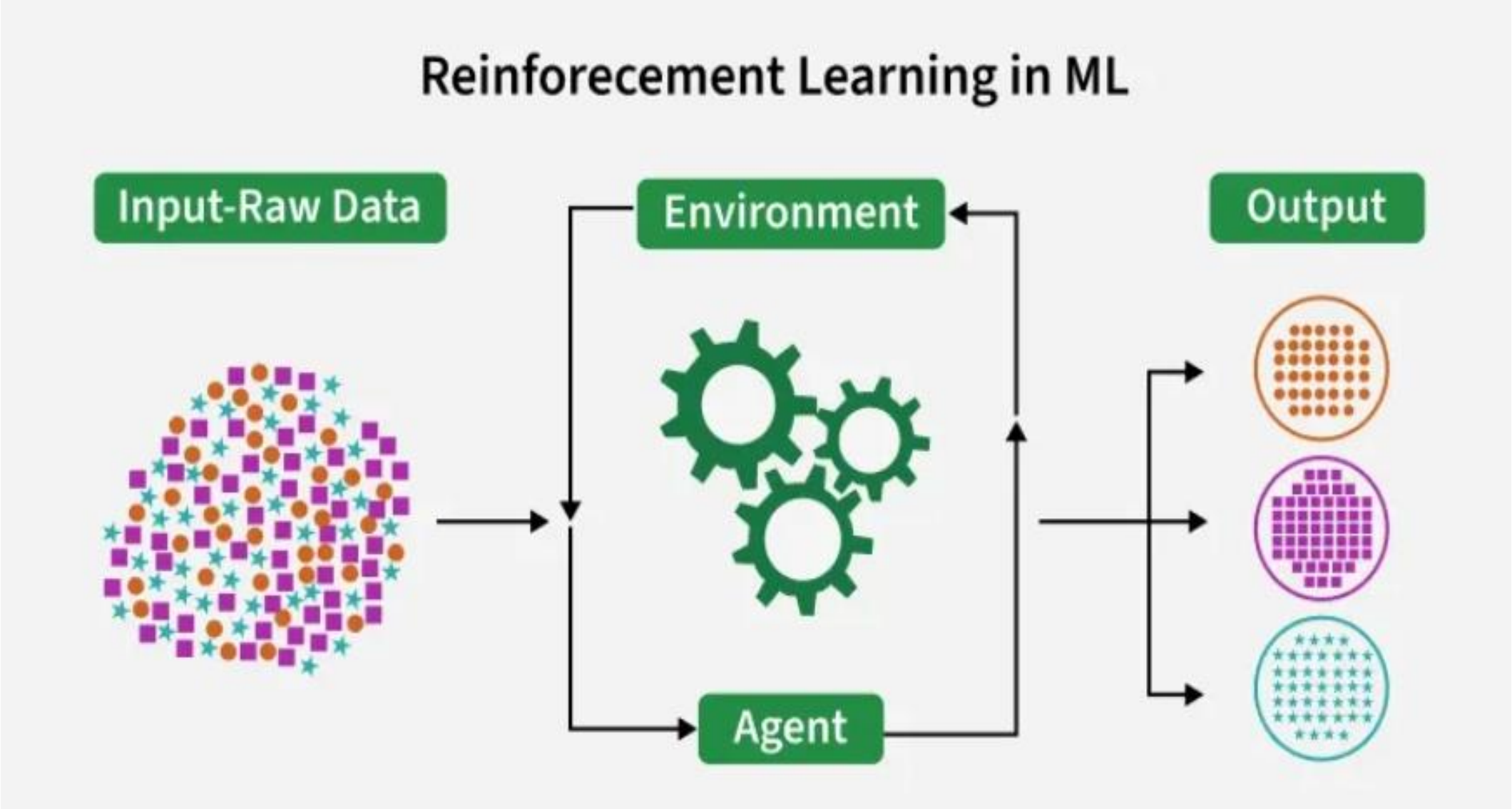

Unsupervised Learning
INPUT
RAW DATA
Output
Interpretation
Algorithm
Processing
• Unknown Output
• No Training Data Set
Model training

- The image shows set of animals like elephants, camels and cows that represents raw data that the unsupervised learning algorithm will process.
- The "Interpretation" stage signifies that the algorithm doesn't have predefined labels or categories for the data. It needs to figure out how to group or organize the data based on inherent patterns.
- An algorithm represents unsupervised learning process that helps to identify patterns in the data.
- The processing stage shows the algorithm working on the data.

The output shows the results of the unsupervised learning process. In this case, the algorithm might have grouped the animals into clusters based on their species (elephants, camels, cows).

Reinforcement Learning

- Reinforcement Learning (RL) is a branch of machine learning that focuses on how agents can learn to make decisions through trial and error to maximize cumulative rewards.
- RL allows machines to learn by interacting with an environment and receiving feedback based on their actions.
- This feedback comes in the form of rewards or penalties.


Reinforecement Learning in ML
Input-Raw Data
Environment
Output
Agent

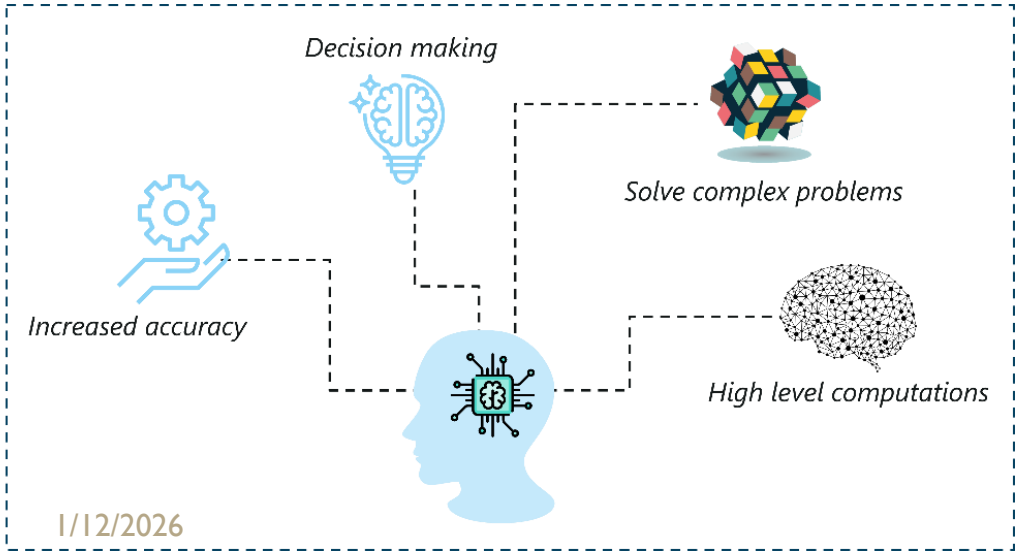
Reinforcement Learning revolves around the idea that an agent (the learner or decision-maker) interacts with an environment to achieve a goal. The agent performs actions and receives feedback to optimize its decision-making over time.

- **Agent**: The decision-maker that performs actions.
- **Environment**: The world or system in which the agent operates.
- **State**: The situation or condition the agent is currently in.
- **Action**: The possible moves or decisions the agent can make.
- **Reward**: The feedback or result from the environment based on the agent's action.

Difference between Machine Learning, Artificial Intelligence, Deep Learning and Data Science

1. Artificial Intelligence (AI)

AI is the broad field of creating machines that can perform tasks that typically require human intelligence. These tasks include reasoning, learning, problem-solving, understanding natural language, and perception.



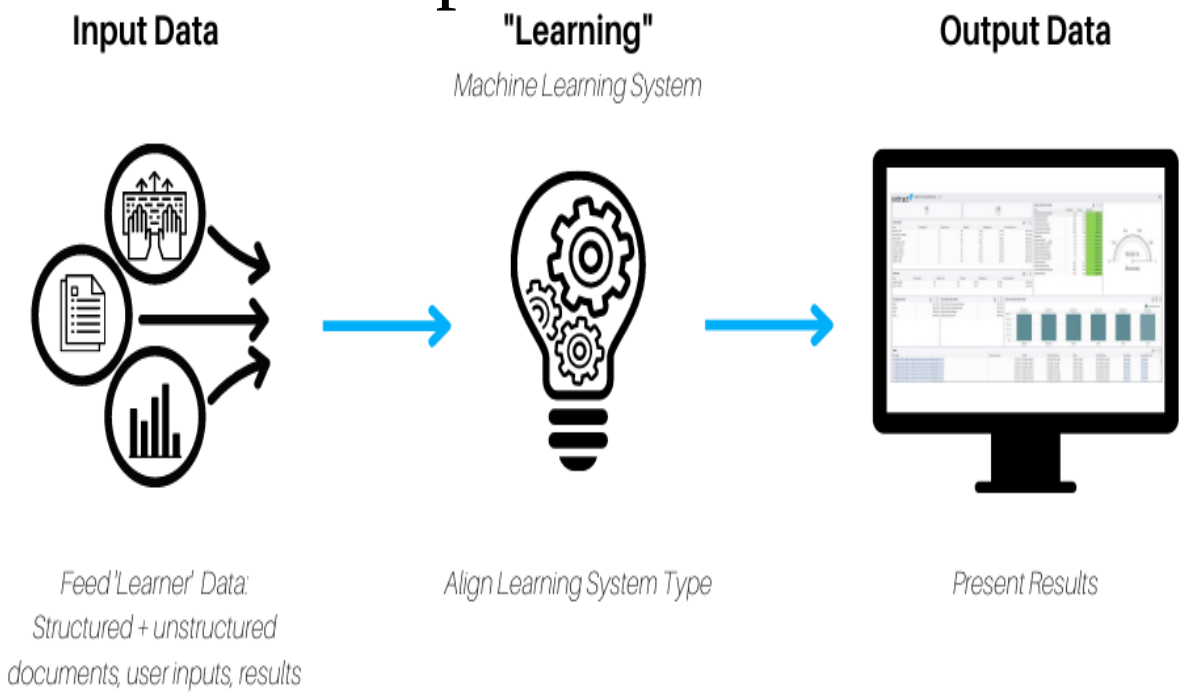
Techniques/Applications:

- Expert systems
- Natural language processing (NLP)
- Robotics
- Computer vision
- Game playing

- Virtual assistants (e.g., Siri, Alexa), recommendation systems (e.g., Netflix, Amazon

2. Machine Learning (ML)

- ML is a subset of AI focused on developing algorithms that allow computers to learn from and make predictions or decisions based on data. It emphasizes the creation of models that can generalize from examples.



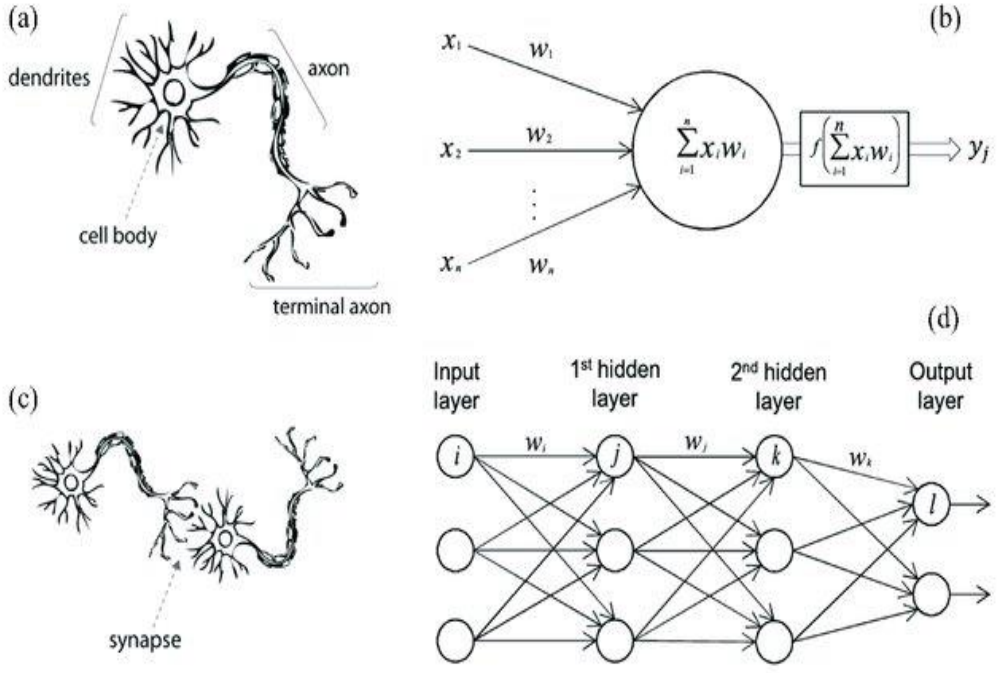
Techniques/Applications:

- Regression (e.g., linear regression)
- Classification (e.g., decision trees, support vector machines)
- Clustering (e.g., k-means clustering)
- Anomaly detection (e.g., fraud detection)

- Spam detection, customer segmentation, predictive maintenance

3. Deep Learning (DL)

DL is a subset of ML that uses neural networks with many layers (deep neural networks) to model complex patterns in data. It is particularly effective in tasks that involve high-dimensional data such as images, audio, and text.



4. Data Science

Data Science is an interdisciplinary field that uses scientific methods, processes, algorithms, and systems to extract knowledge and insights from structured and unstructured

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Programming tools used in Machine Learning

Machine Learning (ML) involves various programming tools and frameworks that facilitate the development, training, and deployment of models. These tools range from programming languages to libraries and integrated development environments (IDEs).

Here are some of the key tools commonly used in ML:

Programming Languages

Python

- Python is the backbone of today's Machine Learning ecosystem. With its simplicity, vast library support and strong community, Python enables rapid prototyping and smooth model development.
- It supports complete end to end ML workflows from data preprocessing to deployment making it ideal for both learners and professionals.

1. Numpy
2. Pandas
3. Keras
8. OpenCV
Python Libraries
4. SciPy
7. Pytorch
6. Scikit-learn
5. Tensorflow

Essential Python Libraries for Machine Learning

- **NumPy**: Provides fast array operations, linear algebra and vectorized computations for scientific computing.
- **Pandas**: Offers DataFrame structures for efficient data cleaning, manipulation and transformation.
- **Matplotlib**: Used to create basic visualizations like line plots, bar charts, histograms and scatter plots.
- **Scikit-learn**: Provides ML algorithms for classification, regression, clustering, dimensionality reduction and evaluation.

- **SciPy**: Extends NumPy with advanced tools for optimization, integration, interpolation and scientific calculations.

- **TensorFlow** and **Keras**: Enables building and training deep learning models with GPU support and production deployment capabilities.
- **PyTorch**: Provides a flexible tensor-based framework with GPU support for building and training neural networks

Setting Up Python for Machine Learning

- Before starting with Machine Learning, you need a proper Python environment.
- There are two commonly used methods to set up Python for ML tasks